

Jafar Isbarov

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Summary

PhD researcher at Virginia Tech working on AI control and security. I build and break safety mechanisms for LLM agents: my recent work demonstrates attacks on CoT monitoring (agent-as-a-proxy) and real-world prompt injection in AI-powered CI/CD pipelines. Published at ACL and NeurIPS, with 3 years of industry experience shipping production NLP systems, including pre-training LLMs from scratch.

Skills

Languages: Python, Julia, C++, Bash, SQL

Cloud and infrastructure: MS Azure, GCP, Debian-based Linux, Docker, multi-GPU jobs

Frameworks: scikit-learn, PyTorch, TensorFlow, Transformers, FastAPI, accelerate

Research: AI control protocols, CoT monitoring, prompt injection attacks, CI/CD security for AI systems, adversarial red-teaming, custom eval harnesses, LLM pre-training

Education

Virginia Tech

Aug. 2025 – Dec. 2028

PhD in Computer Science (GPA: 3.92)

- Demonstrated real-world prompt injection attacks in AI-powered CI/CD pipelines [1].
- Co-developed CASPIAN, the first online detector for cascade attacks in LLM multi-agent systems [2].
- Demonstrated that AI control protocols relying on CoT monitoring can be bypassed via agent-as-a-proxy attacks [3].

The George Washington University

Aug. 2023 – May 2025

MS in Computer Science (GPA: 3.92)

- **Thesis:** Evaluating Language Understanding and World Knowledge of Large Languages Models in Turkic Languages

ASOIU

Aug. 2018 – May 2023

BS in Biomedical Engineering

- **Thesis:** Computational Methods for the Analysis of Single-Cell Transcriptomics Data

Experience

Research Fellow

remote

PRISM AI Safety Fellowship (6% acceptance rate)

Jun. 2026 – Sep. 2026

- Building efficient synchronous monitors for coding agents.
- Red-teaming IFG-based monitors for coding agents.

Machine Learning Engineer

Baku, Azerbaijan

eiLink R&D · Prodata LLC · Azerbaijan Artificial Intelligence Lab

Jul. 2022 – Aug. 2025

- Developed an agentic chatbot for querying heterogeneous drilling datasets for **British Petroleum**, cutting typical analysis and reporting tasks from 2–8 hours to 15–30 minutes (10–20x).
- Built monolingual encoder-only foundation models outperforming multilingual SOTA models for Azerbaijani language understanding by 0.5%. [5]

Research Affiliate

New York, NY

New York University

July 2024 – Dec. 2024

- Worked on an adversarial paradigm for prompt compression with Duygu Ataman.
- Built TUMLU – the first native multilingual language understanding benchmark for Turkic languages. [4]

Publications

[1] **GitInject: Real-World Prompt Injection Attacks in AI-Powered CI/CD Pipelines** Dec. 2026

Jafar Isbarov, Umid Suleymanov, Ilia Shumailov, Murat Kantarcioglu

NeurIPS 2026, Evaluations & Datasets Track ([arXiv](#) [🔗](#)) — Introduces GitInject, an open-source framework for evaluating prompt injection in live GitHub workflows; studies 11 attacks across four AI providers.

[2] **CASPIAN: Online Detection and Attribution of Cascade Attacks in LLM Multi-Agent Systems via Cross-Channel Causal Monitoring** Dec. 2026

Kavana Venkatesh, *Jafar Isbarov*, Saad Amin, Murat Kantarcioglu, Jiaming Cui

NeurIPS 2026 ([arXiv](#) [🔗](#)) — First online framework for detecting cascade attacks in LLM multi-agent systems and attributing them to origin agents, with sub-1% latency overhead.

[3] **Bypassing AI Control Protocols via Agent-as-a-Proxy Attacks** Jan. 2026

Jafar Isbarov, Murat Kantarcioglu

[arXiv](#) [🔗](#) — Introduces agent-as-a-proxy attacks, showing that CoT-monitoring-based control protocols can be bypassed with an X% success rate against monitored agents.

[4] **TUMLU: A Unified and Native Language Understanding Benchmark for Turkic Languages** Aug. 2025

Jafar Isbarov, Arofat Akhundjanova, Mammad Hajili, et al.

[ACL 2025, Main Conference](#) [🔗](#) — First natively-authored (non-translated) benchmark for Turkic languages, covering 8 languages; adopted by [labs/teams] for multilingual evals.

[5] **Open foundation models for Azerbaijani language** Aug. 2024

Jafar Isbarov, Kavsar Huseynova, Elvin Mammadov, et al.

[ACL 2024, 1st SIGTURK Workshop](#) [🔗](#) (**Honorable Mention in the Best Paper Category**)

Awards & Grants

Gemini Academic Program Award — USD 10,000 in Google Cloud credits for Gemini research at Virginia Tech Sep. 2026

Professional Service

Reviewer at:

- ICML 2026
- 2nd SIGTURK Workshop (EMNLP 2026)
- 1st LLMSEC Workshop (ACL 2025)
- ACL ARR 2025 February
- Neural Computing and Applications Journal (2024)
- 4th Multilingual Representation Learning Workshop (EMNLP 2024)

Officer at ACL Special Interest Group in Turkic Languages ([SIGTURK](#) [🔗](#)) (2025)

Extracurricular Activities

Open-source contributions to:

- Hugging Face Ecosystem ([lighteval](#) [🔗](#), [Python Client](#) [🔗](#))
- Julia Ecosystem ([TableTransforms.jl](#) [🔗](#), [Julia-LLM-Leaderboard](#) [🔗](#), [CounterfactualExplanations.jl](#) [🔗](#))
- [OpenBench](#) [🔗](#)